A First Course in Abstract Algebra

Section 41: Simplicial Complexes and Homology Groups

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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41 Simplicial Complexes and Homology Groups

We use integer coefficients and ordinary, unreduced homology. Brackets list the vertices of an oriented simplex; exchanging two vertices reverses its sign. The zero group is denoted by 0.

Exercise 41.1: Boundary of the specified 2-chain

Use $[ijk] = P_iP_jP_k$ with the order indicating orientation. Reversing two vertices changes the sign, so the chain in the question becomes

$$c = 2[134] - 4[346] - 3[234] - [146].$$

(a) Apply $\partial[ijk] = [jk] - [ik] + [ij]$ term by term:

$$\begin{aligned}\partial c &= 2([34] - [14] + [13]) - 4([46] - [36] + [34]) \\ &\quad - 3([34] - [24] + [23]) - ([46] - [16] + [14]) \\ &= 2[13] - 3[14] + [16] - 3[23] + 3[24] \\ &\quad - 5[34] + 4[36] - 5[46].\end{aligned}$$

(b) No: distinct oriented edges form a basis of the chain group, and for instance the coefficient of [46] is $-5 \neq 0$. Thus $\partial c \neq 0$. **(c)** Yes: $\partial(\partial c) = 0$ because the boundary of a boundary is zero. This can also be checked by replacing each $[ij]$ by $P_j - P_i$; every vertex coefficient cancels.

Exercise 41.2: The boundary of a tetrahedron boundary

First

$$\partial_3[1234] = [234] - [134] + [124] - [123].$$

Applying ∂_2 gives

$$\begin{aligned}&([34] - [24] + [23]) - ([34] - [14] + [13]) \\ &\quad + ([24] - [14] + [12]) - ([23] - [13] + [12]) = 0.\end{aligned}$$

Every edge occurs twice with opposite sign. Thus $\partial_2\partial_3[1234] = 0$, as required. The signs, rather than an absence of geometric edges, are what cause the cancellation.

Exercise 41.3: The chain groups of one point

There is exactly one vertex and no positive-dimensional simplex. Hence $C_0(P) = \mathbb{Z}P$ and $C_i(P) = 0$ for $i > 0$. Since ∂_0 maps to zero, $Z_0(P) = C_0(P)$. Since $C_1(P) = 0$, $B_0(P) = 0$. Therefore $H_0(P) \cong \mathbb{Z}$. For $i > 0$, all of C_i, Z_i, B_i, H_i are zero. These are ordinary homology groups; reduced H_0 would instead vanish.

Exercise 41.4: Two isolated points

Here $C_0(X) = \mathbb{Z}P \oplus \mathbb{Z}P'$ and $C_i(X) = 0$ for $i > 0$, because no connecting edge is present. Thus $Z_0 = C_0$, $B_0 = 0$, and $H_0 \cong \mathbb{Z}^2$. All positive-dimensional chain, cycle, boundary, and homology groups are zero. The two independent generators of H_0 record the two connected components; the geometric line segment in the surrounding Euclidean space contributes nothing because it is not part of X.

Exercise 41.5: A closed line segment

Let $e = [12]$. Then $C_1 = \mathbb{Z}e$, $C_0 = \mathbb{Z}P_1 \oplus \mathbb{Z}P_2$, and all C_i for $i \geq 2$ vanish. The boundary satisfies $\partial_1(ne) = n(P_2 - P_1)$. Since the vertices are independent, the boundary is zero only if $n = 0$. Thus $Z_1 = 0$; also $B_1 = 0$, so $H_1 = 0$. In dimension zero, $Z_0 = C_0$ and $B_0 = \mathbb{Z}(P_2 - P_1)$. The sum-of-coefficients map $aP_1 + bP_2 \mapsto a + b$ is onto $\mathbb{Z}$ and has kernel B_0, so $H_0 \cong \mathbb{Z}$. In higher dimensions all groups are zero. This describes each of C_i, Z_i, B_i, H_i explicitly.

Exercise 41.6: True or false: chains, cycles, and boundaries

(a) True, because $\partial^2 = 0$.

(b) False. A once-around cycle of a circle cannot bound a 2-chain in that circle, since the circle has no 2-simplexes.

(c) True. One orientation of each n-simplex is a free abelian basis for $C_n(X)$.

(d) True. $B_n(X)$ is a subgroup of the free abelian group $C_n(X)$, so the subgroup theorem for free abelian groups applies.

(e) True for the same reason, with $Z_n(X) \leq C_n(X)$.

(f) True. $H_n = Z_n/B_n$ is a quotient of an abelian group.

(g) False. The boundary of a 3-simplex is a signed sum of four 2-simplexes, not a single 2-simplex.

(h) True. The boundary of a 2-simplex is a signed sum of edges, hence a 1-chain.

(i) True. A 3-cycle has boundary zero, and zero is a valid 2-chain.

(j) True. Quotienting a group by itself gives the one-element group.

Exercise 41.7: Oriented simplexes in arbitrary dimension

(a) An n-simplex is the convex hull of $n+1$ affinely independent vertices $P_0, \ldots, P_n$. An orientation is a choice of ordering modulo even permutations; an odd permutation reverses the orientation (for $n = 0$ there is just its single vertex). **(b)** Its oriented boundary is

$$\partial_n[P_0 \cdots P_n] = \sum_{i=0}^{n} (-1)^i [P_0 \cdots \widehat{P_i} \cdots P_n],$$

where a hat means omission. For example the formula gives $\partial[P_0P_1] = P_1 - P_0$. **(c)** A codimension-one face is obtained by omitting one vertex, with the induced sign $(-1)^i$ in this boundary formula. More generally a k-face is spanned by any $k+1$ of the vertices. In the textbook's oriented-face terminology, choose its orientation so that it appears with positive sign when it is the specified boundary summand.

Exercise 41.8: Chain, cycle, and boundary groups in general

Use the same definitions with an arbitrary integer subscript n: $C_n(X)$ is the free abelian group on a choice of orientation for every n-simplex; opposite orientations have opposite signs. Define

$\partial_n : C_n \to C_{n-1}$ on each basis simplex by the alternating face formula in Exercise 41.7, and extend by integer linearity. Put $C_n = 0$ for $n < 0$ and $\partial_0 = 0$. Then

$$Z_n(X) = \ker \partial_n, \qquad B_n(X) = \operatorname{im} \partial_{n+1}.$$

Thus a cycle is a chain whose boundary is zero, and a boundary is the boundary of a chain one dimension higher. This gives a uniform answer regardless of how large the dimension is.

Exercise 41.9: Why the boundary squared is zero

By linearity it suffices to start with one oriented n-simplex $[P_0 \cdots P_n]$. Every codimension-two face omits two vertices P_i, P_j with $i < j$. There are exactly two ways for this face to occur in the double boundary. Omitting P_i first moves P_j into position $j-1$, giving sign $(-1)^i(-1)^{j-1}$. Omitting P_j first leaves P_i in position i, giving sign $(-1)^j(-1)^i$. These signs are opposites, so the two copies cancel. Every summand is paired this way, proving $\partial_{n-1}\partial_n = 0$ for $n \geq 2$. For $n = 1$ the conclusion follows from $\partial_0 = 0$.

Exercise 41.10: Coboundaries in the solid tetrahedron

Fix increasing vertex order as the positive orientation. The coefficient of a simplex τ in $\delta\sigma$ is the coefficient of σ in $\partial\tau$; this retains the orientation signs in the question. **(a)** Since $\partial[ij] = P_j - P_i$, we have

$$\delta^{(0)}P_1 = -[12] - [13] - [14], \qquad \delta^{(0)}P_4 = [14] + [24] + [34].$$

(b) The oriented edge $P_3P_2 = -[23]$ occurs with coefficient -1 in the boundaries of $[123]$ and $[234]$, and not in the other two faces. Thus

$$\delta^{(1)}(P_3P_2) = -[123] - [234].$$

(c) Since $P_3P_2P_4 = -[234]$ and $[234]$ has coefficient $+1$ in $\partial[1234]$,

$$\delta^{(2)}(P_3P_2P_4) = -[1234].$$

Each negative sign describes the opposite orientation, not an omitted geometric face.

Exercise 41.11: Defining the coboundary and checking its square

(a) For the finite complexes used here identify $C^{(n)}(X)$ with integer coefficient vectors on the oriented n-simplexes. If D_{n+1} is the boundary matrix, define $\delta^{(n)}$ to have matrix D_{n+1}^{T}. Equivalently, sum the oriented cofaces with their incidence signs and extend linearly. **(b)** Since $D_{n+1}D_{n+2} = 0$, transposing gives

$$D_{n+2}^{\mathsf{T}} D_{n+1}^{\mathsf{T}} = 0.$$

This is precisely $\delta^{(n+1)}\delta^{(n)} = 0$. *Scope qualification.* For an arbitrary infinite complex, cochains should be defined as $\operatorname{Hom}(C_n(X), \mathbb{Z})$, allowing arbitrary coefficient functions; they need not equal the finite-support chain group. Define $(\delta\alpha)(c) = \alpha(\partial c)$ there. The same square-zero proof then works without a finiteness assumption. The textbook's identification of chains and cochains is valid for its finite examples.

Exercise 41.12: Cocycles and coboundaries

Define

$$Z^{(n)}(X) = \ker \delta^{(n)}, \qquad B^{(n)}(X) = \operatorname{im} \delta^{(n-1)}.$$

For $n = 0$ set $B^{(0)} = 0$. Kernels and images of homomorphisms are subgroups. If $b = \delta^{(n-1)}c$, then $\delta^{(n)}b = 0$ because $\delta^2 = 0$. Therefore every coboundary is a cocycle and $B^{(n)}(X) \leq Z^{(n)}(X)$, making the next quotient definition possible.

Exercise 41.13: Cohomology of the tetrahedron surface

Define $H^{(n)}(X) = Z^{(n)}(X)/B^{(n)}(X)$. For the surface S there are four vertices, six edges, four faces, and no 3-simplex. In dimension zero $\delta f = 0$ says that f takes the same value at both ends of every edge, so f is constant. Hence $H^{(0)}(S) \cong \mathbb{Z}$. For dimension one, a cocycle has edge values u_{ij} satisfying $u_{jk} - u_{ik} + u_{ij} = 0$ on every triangular face. Put $t_1 = 0$, $t_j = u_{1j}$; the face $[1ij]$ gives $u_{ij} = t_j - t_i$. Thus every 1-cocycle is the coboundary of a vertex function and $H^{(1)}(S) = 0$. In dimension two every cochain is a cocycle, since there are no 3-simplexes. Write e_{ijk} for the cochain dual to a face. Coboundaries of edges give, among others, the relations

$$e_{123} + e_{124} = 0, \quad -e_{123} + e_{134} = 0, \quad e_{123} + e_{234} = 0.$$

The other edge relations follow from these. Thus all four face classes are signed multiples of one unrestricted generator; there is no torsion relation. Consequently

$$H^{(2)}(S) \cong \mathbb{Z}, \qquad H^{(n)}(S) = 0 \ (n > 2).$$

These are cohomology computations from the coboundary maps, rather than an assumption that cohomology always equals homology.